

Notes: Matsa (JF, 2010)

Capital Structure as a Strategic Variable: Evidence from Collective Bargaining

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1 Executive overview

Standard corporate finance explains leverage through taxes, distress costs, agency, and information. Matsa (2010) emphasizes a complementary channel: **strategic leverage in input markets**. When wages are negotiated with a union, holding more liquid resources can increase workers' wage demands. If the firm cannot easily refinance at the bargaining date, then choosing higher debt ex ante reduces *expropriable liquidity* at the bargaining stage, thereby improving the firm's bargaining position. This strategic benefit is strongest when profits are volatile: volatility increases upside realizations where unions can extract rents, while distress costs matter for both union and nonunion firms. Hence the key testable prediction is an **interaction**: union power $\times$ profit variability.

2 Theoretical framework (Section I)

2.1 Timing and primitives (Figure 1)

The model is a three-date contracting problem:

1. At $t = 0$, the firm chooses financing and invests I using *unsecured* debt and equity. Debt has face value D due at $t = 1$.
2. At $t = 1$, an intermediate cash-flow shock $\tilde{r}$ is realized. To continue, the firm must hire unionized labor and negotiate a wage bill $w(\tilde{r})$ paid out of current cash flow. The debt payment D is also due at $t = 1$.
3. At $t = 2$, conditional on continuation, the project yields a continuation payoff $\tilde{R}$.

Two simplifying frictions are crucial:

- **Non-pledgeability of continuation value:** $\tilde{R}$ cannot be pledged to investors at $t = 0$ or $t = 1$, so it cannot be used to back refinancing at $t = 1$.
- **No refinancing at $t = 1$:** the face value D is taken as a hard commitment when bargaining occurs.

Throughout the theoretical section, investors and the union are risk neutral, and competitive investors require zero expected return.

2.2 Wage determination via generalized Nash bargaining

At $t = 1$, conditional on having cash flow r and outstanding debt D , the firm and the union bargain over the wage bill w . Outside options are normalized to zero (liquidation for the firm; outside wage for workers).

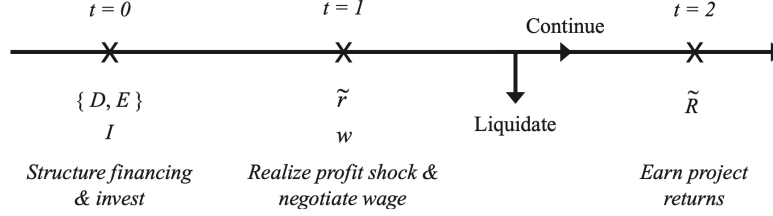


Figure 1. Timing of the model.

Let $z \in [0, 1]$ denote the union's bargaining power, so $1 - z$ is management's bargaining power. The generalized Nash solution chooses w to maximize a product of surpluses. Because the only immediately divisible resource is the firm's *excess liquidity* $r - D$, the bargaining problem reduces to:

$$\max_{0 \leq w \leq r-D} w^z (r - D - w)^{1-z}.$$

Taking logs and applying the first-order condition:

$$\frac{z}{w} - \frac{1-z}{r-D-w} = 0 \implies z(r-D-w) = (1-z)w \implies w(r) = z(r-D).$$

$$w(r) = z(r-D) \tag{1}$$

Interpretation. The union extracts a share z of the *excess liquidity* available at $t = 1$. Higher debt lowers $r - D$ and therefore mechanically reduces the negotiated wage.

2.3 Continuation vs. liquidation

Continuation requires that the firm can pay both the wage and the debt at $t = 1$. Given (1), continuation is feasible when:

$$r - w(r) \geq D \iff r - z(r-D) \geq D \iff (1-z)r \geq (1-z)D \iff r \geq D.$$

Thus the debt face value D pins down a **default/liquidation threshold**.

2.4 Debt payoffs and the value of issuing debt

If $r \geq D$, the firm continues and debt is repaid in full (D). If $r < D$, the firm cannot meet the promised payment and is liquidated; debt holders recover the realized cash flow r . Thus the debt payoff at $t = 1$ is:

$$\min\{r, D\}.$$

With competitive, risk-neutral investors, the $t = 0$ proceeds from issuing face value D equal the expected payoff:

$$V_D(D) = \mathbb{E}[\min\{r, D\}].$$

2.5 Shareholder objective and optimal leverage

Assume r is uniformly distributed on $[\bar{r} - \sigma, \bar{r} + \sigma]$ with $\bar{r} > \sigma$. Then for any $D \in [\bar{r} - \sigma, \bar{r} + \sigma]$,

$$V_D(D) = \int_{\bar{r}-\sigma}^D r \frac{1}{2\sigma} dr + \int_D^{\bar{r}+\sigma} D \frac{1}{2\sigma} dr.$$

Shareholders receive nothing when $r < D$. When $r \geq D$, after paying wage $w(r) = z(r - D)$ and debt D , the firm retains $(1 - z)(r - D)$ at $t = 1$, and it also obtains the (non-pledgeable) continuation value with expectation $\mathbb{E}[\tilde{R}]$ at $t = 2$. Hence expected equity payoff is:

$$V_E(D) = \int_D^{\bar{r}+\sigma} \left((1 - z)(r - D) + \mathbb{E}[\tilde{R}] \right) \frac{1}{2\sigma} dr.$$

Total firm value (and thus the objective) is $V(D) = V_D(D) + V_E(D)$, which is exactly the expression written in the paper:

$$\max_{D \geq \bar{r}-\sigma} \left[\left(\int_{\bar{r}-\sigma}^D r \frac{1}{2\sigma} dr + \int_D^{\bar{r}+\sigma} D \frac{1}{2\sigma} dr \right) + \int_D^{\bar{r}+\sigma} \left((1 - z)(r - D) + \mathbb{E}[\tilde{R}] \right) \frac{1}{2\sigma} dr \right]. \quad (2)$$

2.5.1 Solving for D^* (derivation)

Differentiate $V(D)$ for an interior optimum. Two standard derivative facts are useful:

- $\frac{d}{dD} \mathbb{E}[\min\{r, D\}] = \Pr(r \geq D)$,
- raising D lowers equity payoffs both by (i) shrinking the continuation region and (ii) reducing inframarginal liquidity.

$$\begin{aligned} \frac{dV}{dD} &= \underbrace{\frac{\bar{r} + \sigma - D}{2\sigma}}_{\text{marginal increase in debt proceeds}} - \underbrace{\frac{\mathbb{E}[\tilde{R}]}{2\sigma}}_{\text{lost continuation value at the margin}} - \underbrace{\frac{(1 - z)(\bar{r} + \sigma - D)}{2\sigma}}_{\text{lost inframarginal equity liquidity}} \\ &= \frac{z(\bar{r} + \sigma - D) - \mathbb{E}[\tilde{R}]}{2\sigma}. \end{aligned}$$

Imposing the lower bound $D \geq \bar{r} - \sigma$ gives the closed form:

$$D^* = \max \left\{ \bar{r} - \sigma, \bar{r} + \sigma - \frac{\mathbb{E}[\tilde{R}]}{z} \right\}. \quad (3)$$

2.5.2 Comparative statics and the key interaction prediction

Equation (3) implies two regimes:

- **Low-union-power regime.** If the second term is below the first, the firm chooses $D^* = \bar{r} - \sigma$. This *decreases* in profit variability σ (a standard “avoid distress” response).
- **High-union-power regime.** If $\bar{r} + \sigma - \mathbb{E}[\tilde{R}]/z > \bar{r} - \sigma$, then $D^* = \bar{r} + \sigma - \mathbb{E}[\tilde{R}]/z$. This *increases* in union power z and also *increases* in σ .

The cutoff between regimes comes from:

$$\bar{r} + \sigma - \frac{\mathbb{E}[\tilde{R}]}{z} > \bar{r} - \sigma \iff z > \frac{\mathbb{E}[\tilde{R}]}{2\sigma}.$$

Economic intuition. Volatility raises the probability of very high r states. Only in those inframarginal high- r states does the union extract rents (because rents scale with $r - D$). Therefore volatility amplifies *union rent seeking* more than it amplifies the strategic benefit of “insurance” for nonunion firms. This is why the theory predicts a strong **interaction**: leverage rises with profit variability *more* when union bargaining power is high.

2.6 Testable implications

Mapping the model to data:

- Union bargaining power $\uparrow$ (larger z) $\Rightarrow$ leverage $\uparrow$.
- Profit variability $\uparrow$ (larger σ) $\Rightarrow$ leverage $\downarrow$ for nonunion firms, but *not* for union firms.
- Hence a positive coefficient on **Union** $\times$ **Variability**.
- The effect should be strongest for **near-term debt** (debt due around bargaining) rather than very long-maturity debt.
- Firms may also use other strategic instruments that matter in bargaining (e.g., **inventory buffers** that weaken strike threats).

3 Empirical strategy I: Cross-sectional evidence (Section II)

3.1 Data and measurement (Table I)

The cross-sectional sample pools three years (1977, 1987, 1999) for which firm-level collective-bargaining coverage can be constructed. Key variables:

- **Union coverage** $\in [0, 1]$: fraction of a firm’s workers covered by collective bargaining.
- **Profit variability**: $\text{sd}(\Delta\text{EBITDA})/\text{Assets}$ computed over the prior decade (at least 5 years of data), standardized by its sample standard deviation (so one unit is one standard deviation).
- **Leverage**: total debt scaled by market value (main) and by book value (robustness).
- Standard leverage controls: tangibility, market-to-book, size, modified Altman z -score, ROA.

Reading Table I. Average total debt/market value is about 0.317. Union coverage averages about 0.27 with substantial dispersion (median 0.20). Profit variability is standardized, so its mean is around 0.73 and its standard deviation is 1 by construction.

3.2 Main cross-sectional regression (Equation 4)

The paper estimates a Rajan–Zingales style leverage regression augmented with union terms:

$$\frac{\text{DEBT}_{ijt}}{\text{VALUE}_{ijt}} = \alpha_1 \text{UNION}_{it} + \alpha_2 \text{VARIABILITY}_{it} + \alpha_3 (\text{UNION}_{it} \times \text{VARIABILITY}_{it}) + X_{it}\beta + \omega_{jt} + \varepsilon_{ijt}, \quad (4)$$

where ω_{jt} are two-digit SIC industry-by-year fixed effects and standard errors are clustered at the firm level.

Parameter mapping to theory. $\alpha_1 > 0$ corresponds to $z \uparrow \Rightarrow D^* \uparrow$. $\alpha_2 < 0$ captures the “distress-cost” response for nonunion firms. $\alpha_3 > 0$ is the key strategic-leverage interaction.

Table I
Summary Statistics for Cross-sectional Analysis

The sample consists of 1,676 firm-year observations from 1977, 1987, and 1999. Estimates of union coverage are from Hirsch (1991) and Eschuk (2001), and the financial data are from Compustat. The sample includes all firms with nonmissing observations for the variables shown below. Union coverage is the proportion of a firm’s workers covered by collective bargaining. Profit variability is the standard deviation of the change in earnings before depreciation and amortization, divided by lagged total assets. This variable is then divided by 0.058 (its sample standard deviation). Profit variability is calculated using data from the previous 10-year period, excluding firms with fewer than 5 years of data. Compustat variables are winsorized at 1% tails.

	Mean	Standard Deviation	25th Percentile	Median	75th Percentile
Panel A: Debt					
Total debt/Market value	0.317	0.221	0.141	0.287	0.467
Total debt/Book value	0.261	0.164	0.154	0.252	0.347
Panel B: Key explanatory variables					
Union coverage	0.270	0.266	0.000	0.200	0.500
Profit variability	0.731	1.000	0.262	0.415	0.737
Panel C: Financial control variables					
Fixed assets (%)	0.355	0.185	0.220	0.322	0.461
Market-to-book ratio	1.868	2.398	0.810	1.231	2.014
Log sales (\$ Mil)	6.279	2.005	4.950	6.345	7.743
Modified z -score	2.149	1.633	1.688	2.391	2.957
Return on assets	0.099	0.101	0.070	0.109	0.149

Table II
Cross-sectional Regression of Total Debt on Unionization

The table summarizes the results from regressions of total debt divided by the market value of the firm (divided by assets in column 4) on the fraction of a firm’s workforce covered by collective bargaining, the variability of the firm’s profits, the interaction between those variables, and a set of controls. Profit variability is measured in units of standard deviations of $\text{sd}(\Delta\text{earnings}/\text{assets})$, where earnings is before depreciation and amortization. When uninteracted, the collective bargaining coverage coefficient measures the effect of the law at the mean of profit variability, and the profit variability coefficient measures the effect for nonunionized firms. Controls in all regressions include industry-by-year fixed effects at the two-digit SIC level. Where indicated, controls also include the following financial controls: the proportion of fixed assets, the market-to-book ratio, log sales, modified Altman’s z -score, and return on assets. Standard errors, clustered at the firm level, are reported in parentheses. Compustat variables are winsorized at 1% tails. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
Union coverage	0.102*** (0.025)	0.108*** (0.026)	0.070*** (0.023)	−0.011 (0.017)
Union coverage * Profit variability		0.066* (0.038)	0.130*** (0.038)	0.071** (0.029)
Profit variability		−0.003 (0.007)	−0.041*** (0.012)	−0.022** (0.010)
R^2 (overall)	0.21	0.21	0.37	0.33
R^2 (excluding FEs)	0.04	0.04	0.28	0.23
Industry-year fixed effects	X	X	X	X
Financial controls			X	X
Using book leverage				X

3.3 Table II: Total leverage and the interaction

What Table II does. It estimates (4) with total debt scaled by market value (and a robustness scaling by assets).

Main results.

- **Union coverage is positively related to leverage:** in column (1) a one-unit increase in union coverage (from 0 to 1) is associated with about 0.102 higher debt/market value. Interpreting at the 10pp level: ≈ 1 percentage point higher leverage for 10pp more union coverage.
- **The interaction is positive:** in column (3), $\alpha_3 \simeq 0.130$ and is highly significant. This means the union–leverage relationship is *much stronger* for firms whose profits are more variable.
- **Profit variability lowers leverage for nonunion firms:** in column (3), $\alpha_2 \simeq -0.041$. This matches the idea that volatility increases distress risk and therefore discourages leverage absent bargaining motives.

Book/asset scaling check (column (4)). Scaling debt by assets weakens the direct union effect but preserves a strong positive interaction. This matters because unions can affect firm market value; the interaction’s robustness suggests the core mechanism is not only valuation scaling.

3.4 Table III: Near-term debt (maturity) evidence

Goal. Debt is strategically relevant because it is due at bargaining. Therefore the union/variability channel should appear most strongly in *near-maturity* debt.

What Table III shows. For debt due within 1–5 years (each scaled by market value), the interaction term $\text{UNION} \times \text{VARIABILITY}$ is positive and statistically significant in every horizon, with economically large magnitudes.

Key pattern. Relative to mean near-term debt, the interaction is largest for short horizons (1–2 years), consistent with the model’s date-1 bargaining mechanism:

- 1-year debt: interaction 0.050 vs mean 0.059 (very large relative effect),
- 2-year debt: interaction 0.083 vs mean 0.089,
- longer horizons remain positive but become smaller relative to the growing mean.

Table III

Cross-sectional Regressions of Near-Term Debt on Unionization

The table summarizes the results from regressions of various measures of near-term debt divided by the market value of the firm on the fraction of a firm's workforce covered by collective bargaining, the variability of the firm's profits, the interaction between those variables, and a set of controls. The specifications are the same as that reported in Table II, column 3, but the dependent variable in each column is debt due within the number of years indicated, divided by the market value of the firm. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: Debt Due within X Years /Market Value				
	1 year	2 years	3 years	4 years	5 years
Union coverage	0.012 (0.009)	0.021* (0.012)	0.033** (0.015)	0.041** (0.017)	0.039** (0.019)
Union coverage	0.050*** (0.014)	0.083*** (0.021)	0.095*** (0.024)	0.116*** (0.029)	0.119*** (0.032)
* Profit variability					
Profit variability	-0.009* (0.005)	-0.019*** (0.006)	-0.022*** (0.007)	-0.029*** (0.007)	-0.038*** (0.008)
Observations	1,676	1,498	1,495	1,494	1,484
R ² (overall)	0.17	0.21	0.22	0.27	0.29
R ² (excluding FEs)	0.10	0.13	0.14	0.17	0.18
Mean of dependent variable	0.059	0.089	0.118	0.147	0.171
Industry-year fixed effects	X	X	X	X	X
Financial controls	X	X	X	X	X

Table IV

Cross-sectional Regressions of Inventories on Unionization

The table summarizes the results from regressions of various measures of inventories divided by sales on the fraction of a firm's workforce covered by collective bargaining, the variability of the firm's profits, the interaction between those variables, and a set of controls. The specifications are the same as that reported in Table II, column 3, but with different dependent variables. * and *** denote statistical significance at the 10% and 1% levels, respectively.

	Total Inventories	Raw Materials	Work-in-Progress	Finished Goods
Union coverage	0.017* (0.010)	0.004 (0.005)	-0.004 (0.007)	0.006 (0.007)
Union coverage	0.017 (0.013)	0.002 (0.006)	0.013*** (0.005)	0.013* (0.007)
* Profit variability				
Profit variability	-0.021*** (0.006)	-0.002 (0.004)	-0.010*** (0.003)	-0.006* (0.003)
Observations	1,669	1,103	934	1,030
R ² (overall)	0.48	0.35	0.33	0.24
R ² (excluding FEs)	0.24	0.10	0.08	0.08
Mean of dependent variable	0.158	0.058	0.048	0.071
Industry-year fixed effects	X	X	X	X
Financial controls	X	X	X	X

3.5 Table IV: Inventory policy as a complementary strategic variable

Idea. Inventories (especially finished goods / work-in-progress) can weaken the union's strike threat: buffer stock allows continued sales/operations, reducing the union's bargaining leverage.

What Table IV shows. Inventory-to-sales ratios are higher at unionized firms, and the **union × variability** interaction is particularly pronounced for work-in-progress and finished goods inventories.

Interpretation. Profit variability is negatively related to inventories (consistent with tighter liquidity/financing constraints for volatile firms), but this negative relation is weaker among unionized firms: when the bargaining stakes are high, firms appear willing to hold larger strategic inventories despite financing frictions.

4 Empirical strategy II: Labor law quasi-experiments (Section III)

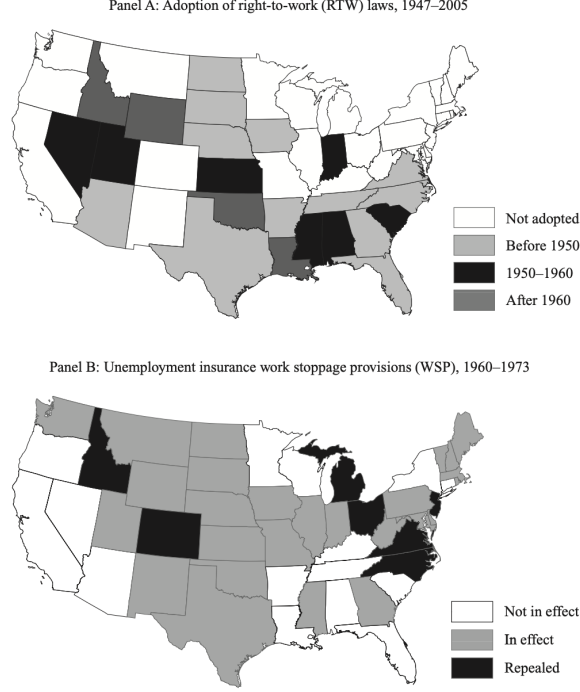
4.1 Motivation for quasi-experimental variation

Cross-sectional union coverage is potentially endogenous: unions may sort into firms/industries with specific unobserved characteristics. To obtain more credible causal evidence, the paper uses state labor-law changes that shift union bargaining power.

4.2 Institutional shocks and Figure 2

Two legal changes are exploited:

- **Right-to-work (RTW) laws** (mainly adopted in the 1950s sample). RTW weakens unions by eliminating union-security arrangements that require workers to join/pay



Source: Ellwood and Fine (1987), Hutchens, Lipsky, and Stern (1989).

Figure 2. Legislative history of selected state labor laws. Panel A shows the pattern of adoption of RTW laws, and Panel B shows the pattern of repeal of WSPs. The Indiana RTW law was later repealed, and the New Jersey WSP was readopted soon after it was abolished.

dues.

- **Unemployment insurance work stoppage provisions (WSP).** In states with WSPs, strikers can qualify for unemployment benefits under certain conditions; repealing WSPs raises the cost of striking and weakens unions.

Figure 2 summarizes the timing and cross-state variation in these laws.

4.3 Panel difference-in-differences model (Equation 5)

For firms in industries with high union presence, the paper estimates:

$$\frac{\text{DEBT}_{ijst}}{\text{VALUE}_{ijst}} = \alpha'_1 \text{LAW}_{st} + \alpha'_2 (\text{LAW}_{st} \times \text{VARIABILITY}_i) + \eta_i + \tau_{jt} + \xi_{ijst}, \quad (5)$$

where:

- η_i are firm fixed effects,
- τ_{jt} are industry-by-year fixed effects,
- standard errors are clustered at the state level.

Sign prediction. Here LAW_{st} is defined so that it *reduces union power* (RTW in effect, or no WSP in effect). Thus the strategic channel predicts $\alpha'_1 < 0$ and $\alpha'_2 < 0$: when union power falls, firms (especially volatile ones) reduce leverage.

4.4 Table V: samples and magnitudes

Table V reports two samples:

- RTW sample (1950–1960): fewer firms/observations, and a small fraction of treated-state observations.
- WSP sample (1960–1973): much larger coverage in Compustat and a larger treated share, yielding more precision.

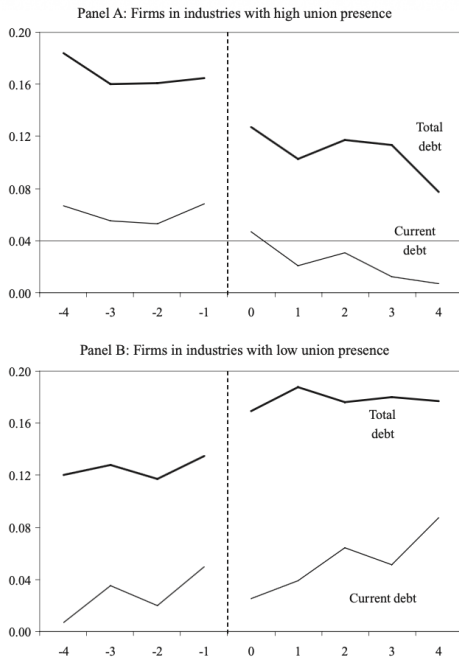


Figure 3. Debt around right-to-work (RTW) law adoption, 1950–1960. This figure depicts average debt divided by total assets in the 4 years before and after the adoption of RTW laws. Panel A includes firms in densely unionized industries (listed in Table AI), and Panel B includes firms in industries with low union presence.

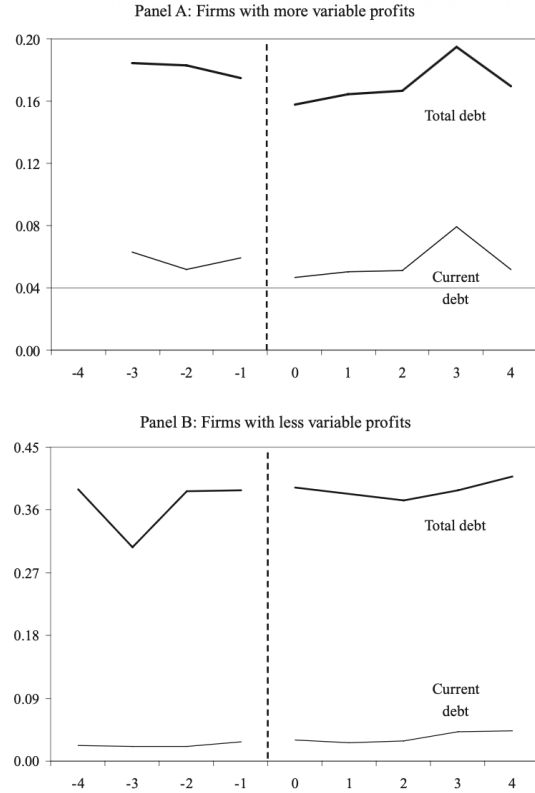


Figure 4. Debt around unemployment insurance work stoppage provision (WSP) repeal, 1960–1973. This figure depicts average debt divided by total assets in the 4 years before and after the repeal of WSPs. Both panels include only firms in densely unionized industries (listed in Table AI). Panel A includes firms with profit variability in the top quartile, and Panel B includes firms with profit variability in the bottom quartile.

4.5 Figures 3–4: event-time visual evidence

Figure 3 (RTW adoption). For firms in highly unionized industries, both total and current debt (scaled by assets) decline after RTW adoption, with little evidence of a pre-

trend. In low-union industries, the same decline is not visible, supporting a union-power interpretation.

Figure 4 (WSP repeal). Among highly unionized industries, firms with *high* profit variability reduce total and current debt after WSP repeal, while low-variability firms show little change. This is a direct visual counterpart to the theory’s interaction prediction.

Table V
Summary Statistics for Labor Law Analysis

This table describes two samples of firms over different time periods: (i) 1950 through 1960, and (ii) 1960 through 1972. The samples consist of all firms in Compustat that have nonmissing observations for debt, total assets, and market value; have at least 5 years of data to calculate profit variability; and are in industries with high union coverage (listed in Table AI). Means are presented with standard deviations in parentheses. Median total assets are in brackets. The number of observations listed is for the debt variables; the financial controls are available for fewer observations.

	Right-to-Work Laws, 1950–1960	Unemployment Insurance Work Stoppage Provisions, 1960–1973
Panel A: Debt		
Total debt/Market value	0.195 (0.181)	0.306 (0.215)
Total debt/Book value	0.157 (0.137)	0.288 (0.183)
Current debt/Market value	0.048 (0.089)	0.061 (0.086)
Current debt/Book value	0.036 (0.064)	0.055 (0.070)
Panel B: Key explanatory and other variables		
States adopting/repealing law	7	7
% sample in adopting states	2.3	21.4
Profit variability	1.574 (1.000)	1.004 (1.000)
Assets (\$ Mil 1999)	2,003 (6,879) [487]	1,936 (7,578) [397]
Panel C: Financial control variables		
Fixed assets (%)	0.373 (0.184)	0.477 (0.268)
Log sales (\$ Mil)	4.815 (1.422)	4.493 (1.554)
ROA	0.123 (0.055)	0.107 (0.060)
Observations	3,277	14,157
Firms	326	1,274

Table VI
Panel Regression Estimates of the Effect of Changes in Labor Law on Total Debt

The table summarizes the results from panel regressions of total debt divided by the market value of the firm (divided by assets in column 3) on a right-to-work (RTW) law or a work stoppage provision (WSP) indicator variable, the interaction between that indicator and the variability of the firm’s profits, and a set of controls. A profit variability main effect is not reported, because it is absorbed by firm fixed effects. Profit variability is measured in units of standard deviations of $sd(\Delta \text{earnings})/\text{assets}$, where earnings is before depreciation and amortization. When uninteracted, the law indicator measures the effect of the law for a firm with mean profit variability. Controls in all regressions include firm and industry-by-year fixed effects. Where indicated, controls also include the following financial controls: the proportion of fixed assets, log sales, and return on assets. In column 4, controls also include the firm’s average (pre-period) operating profits, which is before interest expense, taxes, depreciation, and amortization; is divided by assets; and is normalized by its sample standard deviation. Industry fixed effects are at the two-digit SIC level. The sample includes firms in industries with high union coverage (listed in Table AI). Standard errors, clustered at the state level, are reported in parentheses. Compustat variables are winsorized at 1% tails. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
Panel A: Right-to-work laws, 1950–1960				
RTW law in effect	−0.107** (0.053)	−0.108** (0.052)	−0.077*** (0.024)	−0.109** (0.044)
RTW law in effect * Profit variability	−0.030 (0.026)	−0.037 (0.024)	−0.013 (0.009)	−0.037** (0.019)
RTW law in effect * Average profits				0.007 (0.025)
Observations	3,277	2,976	2,976	2,494
R ² (overall)	0.78	0.82	0.83	0.82
R ² (excluding FEs)	0.01	0.21	0.14	0.23
Panel B: Work stoppage provisions, 1960–1973				
No WSP in effect	−0.013* (0.007)	−0.006 (0.008)	−0.004 (0.007)	−0.008 (0.010)
No WSP in effect * Profit variability	−0.030*** (0.007)	−0.026*** (0.008)	−0.013** (0.006)	−0.026*** (0.006)
No WSP in effect * Average profits				−0.020*** (0.005)
Observations	14,157	13,711	13,711	8,087
R ² (overall)	0.80	0.84	0.86	0.85
R ² (excluding FEs)	0.01	0.39	0.39	0.41
Firm fixed effects	X	X	X	X
Industry-year fixed effects	X	X	X	X
Financial controls		X	X	X
Using book leverage			X	

4.6 Table VI: total debt responses to labor-law changes

RTW laws (Panel A). Debt/market value falls by about 10–11 percentage points after RTW adoption (a very large effect relative to the RTW sample mean). The interaction with variability is negative in all specifications and becomes statistically stronger when additional controls are included.

WSP repeal (Panel B). The main effect of “No WSP” is negative, and the interaction with variability is strongly negative and highly significant. Thus, when union power is reduced, the decline in leverage is concentrated among firms with more volatile profits—the ones with the strongest strategic incentives under the model.

4.7 Table VII: current (short-maturity) debt

Table VII mirrors Table VI but for current debt. The interaction effects are again negative and economically large, and they are particularly pronounced for the RTW analysis. This reinforces the key mechanism: **debt due around bargaining is the most strategic**.

4.8 Table VIII: robustness to alternative variability measures

A concern is that unions (or labor laws) may themselves affect profit variability. Table VIII interacts the law indicator with alternative variability measures designed to reduce direct labor-cost contamination: pre-law variability, sales variability, and profits excluding labor costs.

Conclusion from Table VIII. Across these alternatives, the interaction effects remain negative and often highly significant, supporting the view that the interaction is not an artifact of variability being mechanically shaped by unions.

Table VII
Panel Regression Estimates of the Effect of Changes in Labor Law on Current Debt

The table summarizes the results from panel regressions of current debt divided by the market value of the firm (divided by assets in column 4) on a right-to-work (RTW) law or a work stoppage provision (WSP) indicator variable, the interaction between that indicator and the variability of the firm's profits, and a set of controls. The specifications are the same as those reported in Table VI, but for the different dependent variable. * and *** denote statistical significance at the 10% and 1% levels, respectively.

	(1)	(2)	(3)	(4)
Panel A: Right-to-work laws, 1950–1960				
RTW law in effect	−0.045 (0.036)	−0.050 (0.041)	−0.041 (0.030)	−0.047* (0.027)
RTW law in effect * Profit variability	−0.048*** (0.010)	−0.056*** (0.011)	−0.034*** (0.006)	−0.053*** (0.009)
RTW law in effect * Average profits				−0.006 (0.041)
Observations	3,277	2,976	2,976	2,494
R ² (overall)	0.66	0.69	0.73	0.70
R ² (excluding FEs)	0.00	0.14	0.12	0.15
Panel B: Work stoppage provisions, 1960–1973				
No WSP in effect	−0.006 (0.005)	−0.003 (0.007)	−0.003 (0.005)	−0.002 (0.007)
No WSP in effect * Profit variability	−0.013*** (0.005)	−0.011 (0.008)	−0.009* (0.005)	−0.010 (0.007)
No WSP in effect * Average profits				−0.002 (0.006)
Observations	14,157	13,711	13,711	8,087
R ² (overall)	0.63	0.66	0.66	0.63
R ² (excluding FEs)	0.01	0.20	0.15	0.19
Firm fixed effects	X	X	X	X
Industry-year fixed effects	X	X	X	X
Financial controls		X	X	X
Using book leverage			X	

Table IX
Falsification Test: Estimates from Industries with Low Union Presence

The estimates in this table present a falsification test. **The regression specifications are the same as the ones reported in column 2 of Tables VI and VII, but the sample includes observations of firms in industries with low rates of union coverage** (less than 25% of the workforce covered by collective bargaining). Firms in these industries are unlikely to be directly affected by changes in labor law. ** and *** denote statistical significance at the 5% and 1% levels, respectively.

	Total Debt/ Market Value	Current Debt/ Market Value
Panel A: Right-to-work (RTW) Laws, 1950–1960		
RTW law in effect	0.052** (0.024)	−0.009 (0.024)
RTW law in effect * Profit variability	−0.185*** (0.054)	0.051 (0.067)
Observations	2,154	2,154
R ² (overall)	0.85	0.79
R ² (excluding FEs)	0.34	0.29
Panel B: Work stoppage provisions (WSP), 1960–1973		
No WSP in effect	0.008 (0.011)	0.002 (0.004)
No WSP in effect * Profit variability	0.017** (0.007)	0.011*** (0.003)
Observations	11,512	11,512
R ² (overall)	0.83	0.79
R ² (excluding FEs)	0.34	0.28
Firm fixed effects	X	X
Industry-year fixed effects	X	X
Financial controls	X	X

4.9 Table IX: falsification in low-union industries

Table IX repeats the labor-law DID specifications in industries with low union presence. These firms should be much less affected by RTW/WSP changes through collective bargaining. The estimated patterns differ sharply from the high-union-industry results, supporting the interpretation that the main findings are not driven by generic state-level macro shocks.

Overall, Matsa (2010) provides direct evidence that capital structure is a strategic variable. Firms exposed to stronger union bargaining power use more leverage, and the

Table VIII
Robustness Tests: Estimates Using Alternative Measures of Profit Variability

Reported coefficients are estimated from regressions similar to those reported in Tables VI and VII, column 2, but the law indicator variable is interacted with alternative measures of variability. Except in columns 4 and 8, each variability measure is in units of standard deviations of $\Delta x / \text{assets}$ for different financial variables x . Columns 1 and 5 calculate profit variability only using observations from before the law was changed. Columns 2 and 6 use the variability of sales. Columns 3 and 7 use the variability of profits excluding total labor and related expenses in Panel A, and also excluding pension and retirement expenses in Panel B. In columns 4 and 8, profit variability is measured in units of standard deviations of $\text{sd}(\text{earnings}/\text{assets}_{t-1})$, where earnings is before depreciation and amortization. In all cases, each variability observation is calculated based on at least 5 years of data. ** and *** denote statistical significance at the 5% and 1% levels, respectively.

	Total Debt/Market Value				Current Debt/Market Value			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Right-to-work (RTW) laws, 1950–1960								
RTW law in effect	−0.089 (0.064)	−0.099** (0.047)	0.000 (0.041)	−0.113** (0.051)	−0.062 (0.068)	−0.041 (0.041)	0.047 (0.030)	−0.057 (0.037)
<u>* Profit variability before law</u>	−0.036** (0.017)				−0.031** (0.015)			
RTW law in effect		−0.028*** (0.010)				−0.035*** (0.006)		
<u>* Sales variability</u>								
RTW law in effect			−0.071*** (0.006)				−0.068*** (0.003)	
<u>* Profit variability (excl. labor cost)</u>								
RTW law in effect				−0.065 (0.051)				−0.100*** (0.034)
<u>* Profit level variability</u>								
Observations	2,474	2,973	1,286	2,956	2,474	2,973	1,286	2,956
R^2 (overall)	0.82	0.82	0.83	0.82	0.70	0.69	0.75	0.69
R^2 (excluding FEs)	0.23	0.21	0.25	0.21	0.15	0.14	0.14	0.14
Panel B: Work stoppage provisions (WSP), 1960–1973								
No WSP in effect	0.006 (0.017)	−0.004 (0.007)	0.021 (0.016)	−0.004 (0.007)	0.004 (0.008)	−0.003 (0.007)	0.015 (0.009)	−0.002 (0.006)
No WSP in effect	−0.029*** (0.009)				−0.006 (0.005)			
<u>* Profit variability before law</u>								
No WSP in effect		−0.021 (0.017)				−0.018*** (0.006)		
<u>* Sales variability</u>								
No WSP in effect			−0.023*** (0.009)				0.001 (0.006)	
<u>* Profit variability (excl. labor cost)</u>								
No WSP in effect				−0.017*** (0.004)				−0.012*** (0.004)
<u>* Profit level variability</u>								
Observations	6,425	13,711	2,807	13,711	6,425	13,711	2,807	13,711
R^2 (overall)	0.85	0.84	0.85	0.84	0.65	0.66	0.62	0.66
R^2 (excluding FEs)	0.42	0.39	0.41	0.39	0.20	0.20	0.20	0.20
Firm fixed effects	X	X	X	X	X	X	X	X
Industry-year fixed effects	X	X	X	X	X	X	X	X
Financial controls	X	X	X	X	X	X	X	X

sensitivity of leverage to union power is strongest when profits are more volatile. Quasi-experimental evidence from labor-law changes supports a causal interpretation. A central welfare implication is that bargaining can induce firms to adopt *excessive debt*, increasing vulnerability to negative shocks and potentially distorting investment and competition.

5 Conclusion

5.1 Summary

This paper concludes that **capital structure is not determined only by “pure finance” trade-offs** (taxes, distress, agency), but can also be chosen strategically to influence outcomes in **input-market bargaining**. When a firm bargains with a union, liquidity becomes an object that the union can partially extract. By committing to higher debt ex ante, the firm reduces the cash that can be redistributed to workers at the bargaining stage, thus improving its bargaining position.

The paper makes a strong causal case for this mechanism by combining:

- (i) a simple bargaining model that delivers sharp comparative statics,
- (ii) cross-sectional evidence using firm-level collective bargaining coverage, and

- (iii) quasi-experimental evidence using **labor-law changes** (RTW adoption and WSP repeal) that weaken unions.

The headline conclusion is:

Collective bargaining increases leverage, and the effect is strongest for firms with more volatile profits.

This interaction pattern is precisely what the model predicts when leverage is used strategically.

5.2 Key empirical takeaways

The empirical results can be summarized as three linked facts:

- (1) **Cross-section: union coverage raises leverage, especially when profit variability is high.** In the main cross-sectional specification (Table II, column (3)): union coverage has a positive coefficient (0.070***), profit variability has a negative coefficient for nonunion firms (−0.041***), and the interaction is strongly positive (0.130***). Economically, this implies that for high-variability firms, unionization is associated with much higher leverage than for low-variability firms.
- (2) **Debt maturity: the union $\times$ variability effect is largest for near-term debt.** In Table III, the interaction coefficients for debt due within 1–2 years (0.050*** and 0.083***) are very large relative to the means of those dependent variables, consistent with the model’s key timing idea: **debt matters most when it comes due around bargaining.**
- (3) **Quasi-experimental evidence: weakening unions reduces leverage, especially for volatile firms.** In high-union industries, when labor-law changes weaken unions:
 - RTW adoption reduces total leverage (Table VI, Panel A: about −0.107** in column (1)), and the interaction with variability is negative (e.g., −0.037** in column (4)).
 - WSP repeal (“No WSP in effect”) yields a strongly negative variability interaction (Table VI, Panel B: about −0.030*** in column (1)).

Moreover, the “current debt” regressions (Table VII) show particularly strong negative interaction effects for RTW laws, consistent with strategic effects operating through short-maturity obligations.

5.3 Why debt is strategic in union bargaining

The mechanism is a simple chain:

Step 1: bargaining splits *excess liquidity*. In the model, the union’s wage bill is determined by generalized Nash bargaining. With union bargaining power z , the union extracts a share of the firm’s cash available after honoring debt due at bargaining:

$$w(r) = z(r - D).$$

Higher debt D mechanically reduces $r - D$, shrinking the “pie” the union can target.

Step 2: debt is a commitment device because the firm cannot refinance against future profits. If future continuation value cannot be pledged (or refinancing is limited), then promised payments D are credible commitments at the bargaining date. Debt therefore works like a **hard claim** on cash flows that reduces the resources exposed to rent extraction.

Step 3: volatility strengthens the strategic benefit of debt. Higher profit variability increases the mass of “good” states where there is a lot of liquidity to extract. In those states, unions can demand higher wages; this amplifies the ex ante incentive to pre-commit by issuing more debt. Hence the sharp prediction: **Union power $\times$ Profit variability** should raise leverage.

Step 4: the maturity structure is part of the mechanism. Because the bargaining happens at a particular time, the most effective commitment is debt that is relevant at that time: **near-term debt** (and current debt) is the tightest constraint on available liquidity at bargaining, so the interaction effect should be strongest there (exactly as shown in Table III and Table VII).

Step 5: firms also use non-financial strategic instruments. The paper’s inventory evidence is consistent with the same bargaining logic: inventory buffers can weaken strike threats and reduce the union’s effective leverage. Thus, strategic responses appear in both **financial policy** (leverage and maturity) and **operating policy** (inventory).

5.4 Broader implication

The paper emphasizes that if collective bargaining induces firms to distort capital structure, then labor-market outcomes can be inefficient *even with efficient bargaining*. Strategic leverage can move firms toward **excess debt**, increasing vulnerability to negative cash-flow shocks and potentially distorting investment and product-market competition (via debt overhang and strategic interactions).

5.5 Takeaway

Firms use debt strategically in collective bargaining: higher union power leads to higher leverage, and the effect is strongest for firms with more volatile profits and for near-term debt, with causal support from labor-law shocks that weaken unions and reduce leverage.